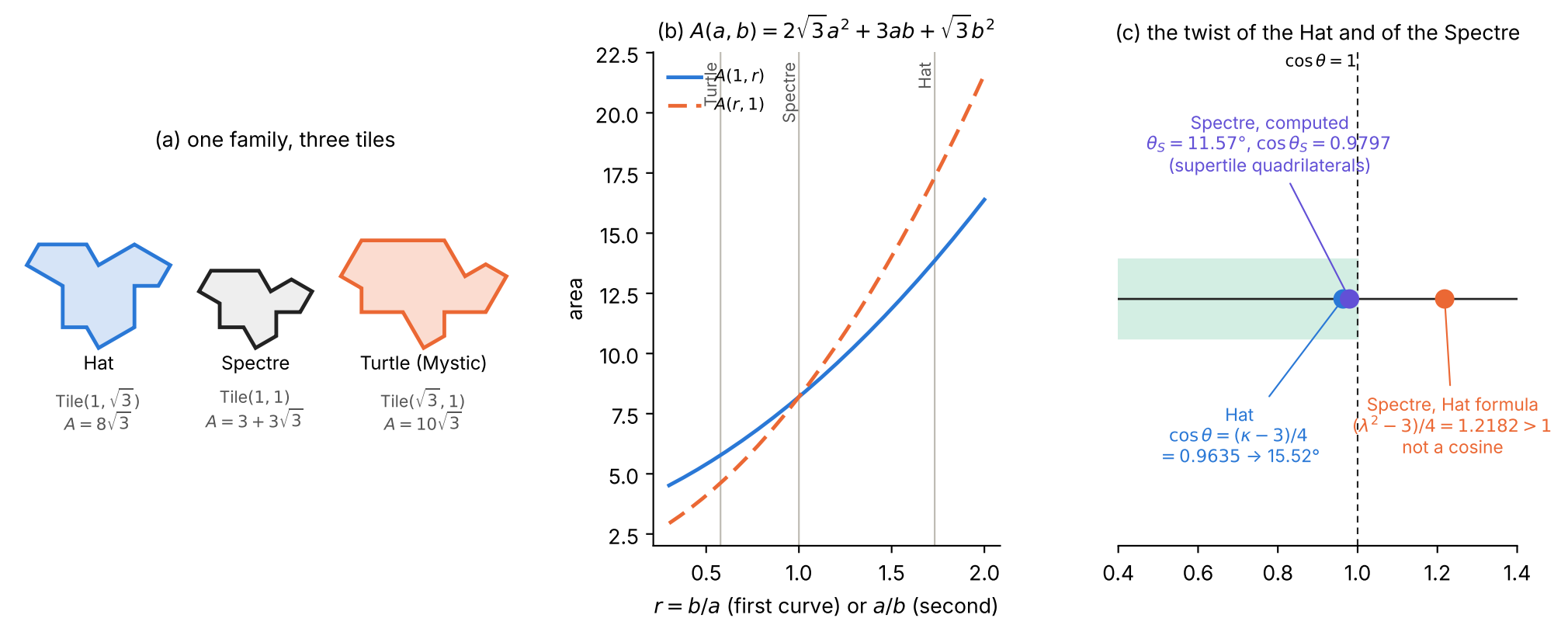
(a) one family, three tiles
Hat
Tile(1, √3)
A = 8√3
Spectre
Tile(1, 1)
A = 3 + 3√3
Turtle (Mystic)
Tile(√3, 1)
A = 10√3
(b) A(a, b) = 2√3 a² + 3ab + √3 b²
A(1, r)
A(r, 1)
Turtle
Spectre
Hat
area
r = b/a (first curve) or a/b (second)
(c) the twist of the Hat and of the Spectre
cos θ = 1
Spectre, computed
θ_S = 11.57°, cos θ_S = 0.9797
(supertile quadrilaterals)
Hat
cos θ = (κ − 3)/4
= 0.9635 → 15.52°
Spectre, Hat formula
(λ² − 3)/4 = 1.2182 > 1
not a cosine